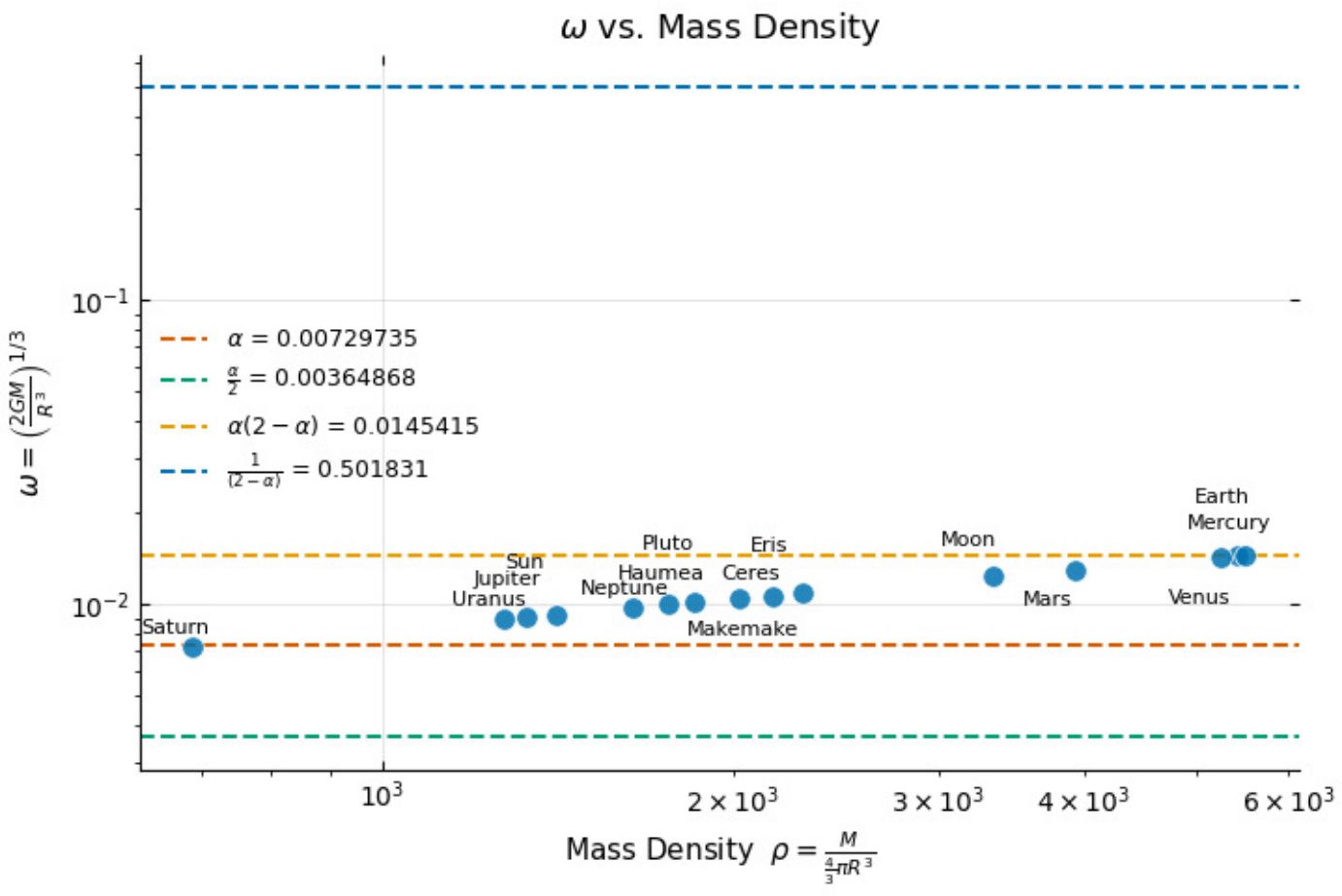

ω vs. Mass Density
$\omega = \left(\frac{2GM}{R^3}\right)^{1/3}$
$10^{-1}$
$10^{-2}$
$\alpha = 0.00729735$
$\frac{\alpha}{2} = 0.00364868$
$\alpha(2-\alpha) = 0.0145415$
$\frac{1}{(2-\alpha)} = 0.501831$
Saturn
Sun
Jupiter
Uranus
Neptune
Pluto
Haumea
Eris
Ceres
Makemake
Moon
Mars
Earth
Mercury
Venus
$10^3$
$2 \times 10^3$
$3 \times 10^3$
$4 \times 10^3$
$6 \times 10^3$
Mass Density $\rho = \frac{M}{\frac{4}{3}\pi R^3}$